

CSc 8830 Computer Vision, Module 3

Image Blurring by Filtering: Spatial Domain vs Fourier Domain

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GitHub repository: <https://github.com/siri423/Module3-ImageFiltering>

1 Objective

There are two parts to this assignment. First, implement image blurring using a filtering approach, meaning I convolve the image with a small blur kernel and write that convolution myself. Second, show that a spatial filter gives the same result as its Fourier domain version, that is, that convolution in space is the same as multiplication in frequency, and use my own implementation as the evidence for it.

Everything runs as one Streamlit web app (`app.py`) where each part is reachable from a tab, with the actual work done in a small set of commented Python files under `src/`.

2 Implementation: blurring by filtering

Blurring an image means replacing every pixel with a weighted average of the pixels around it. The weights are held in a small grid called a kernel. Sliding that kernel across the image and taking the weighted sum at each spot is a 2D convolution:

$$(f * g)(x, y) = \sum_s \sum_t f(s, t) g(x - s, y - t), \quad (1)$$

where f is the image and g is the kernel. I built two kernels, both scaled so their weights add up to 1 (that keeps the brightness of the picture the same):

- A box kernel, where every weight is equal, so it is a plain average of the neighbours.
- A Gaussian kernel, where the weights follow a bell curve, so the centre pixel counts most and the effect fades towards the edges. This gives a smoother, more natural looking blur:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/(2\sigma^2)}. \quad (2)$$

2.1 The spatial convolution

The convolution is in `src/spatial.py` and does not use any library blur function. There is a plain nested loop version that states the definition directly, and a faster version that gives the same answer by looping over the few kernel positions instead of over every pixel:

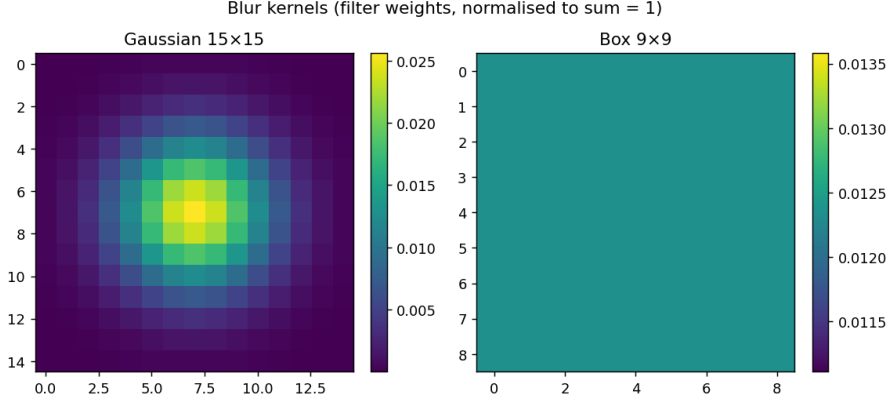


Figure 1: The two blur kernels (the filter weights). The Gaussian keeps most of its weight in the centre, and the box spreads it out evenly.

```
for i in range(kh):
    for j in range(kw):
        # padded[i:i+H, j:j+W] is the image shifted by (i, j)
        out += kflip[i, j] * padded[i:i+H, j:j+W]
```

The kernel is flipped first so this is true convolution rather than correlation (for a symmetric blur kernel the two are the same). Image borders are handled by zero padding, which I use for the proof because it makes the operation a linear convolution, or by mirroring the edge for a nicer looking plain blur.

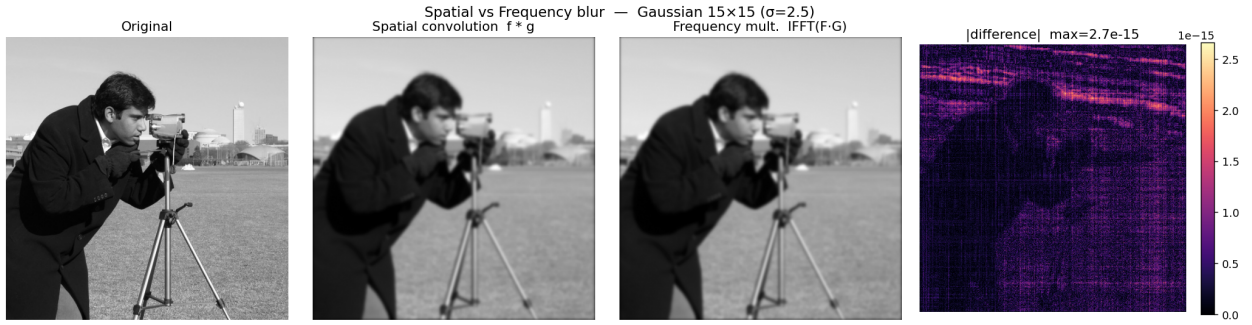


Figure 2: The cameraman test image blurred with a Gaussian 15×15 kernel. Panels 2 and 3 are the two methods from Section 3, and panel 4 is their difference, discussed in Section 5.

3 Theory: the Convolution Theorem

The 2D Discrete Fourier Transform (DFT) of an $M \times N$ image is

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}. \quad (3)$$

The Convolution Theorem says that convolution in one domain matches element-wise multiplication in the other:

$$f * g \iff F \cdot G \quad (4)$$

So the blur can be done either by sliding the kernel (Section 2) or entirely in the frequency domain:

$$f * g = \mathcal{F}^{-1}\{\mathcal{F}\{f\} \cdot \mathcal{F}\{g\}\}. \quad (5)$$

This is what `src/frequency.py` does: take the FFT of the image, take the FFT of the kernel, multiply the two spectra, and take the inverse FFT.

3.1 Proof

It is enough to prove the 1D case, since the 2D transform is just the 1D transform applied along the rows and then the columns. Write $\omega = e^{-j2\pi/N}$ and let $y[n] = \sum_m f[m] g[(n - m) \bmod N]$ be the circular convolution. Its DFT is

$$Y[k] = \sum_{n=0}^{N-1} \left(\sum_{m=0}^{N-1} f[m] g[(n - m) \bmod N] \right) \omega^{nk}. \quad (6)$$

Substitute $p = (n - m) \bmod N$, so $n = m + p$, and because $\omega^N = 1$ we have $\omega^{nk} = \omega^{mk} \omega^{pk}$. The double sum then splits into two:

$$Y[k] = \left(\sum_m f[m] \omega^{mk} \right) \left(\sum_p g[p] \omega^{pk} \right) = F[k] G[k]. \quad (7)$$

So the transform of a convolution is the product of the transforms. The other direction, multiplication in space matching convolution in frequency, follows the same way using the inverse transform.

3.2 Linear vs circular convolution

The plain FFT gives a circular convolution, where the kernel wraps around the image edges. To reproduce the ordinary zero padded (linear) convolution from Section 2, both arrays are zero padded to size $(M + k - 1) \times (N + k - 1)$ before transforming (with k the kernel width), and the middle is cropped out afterwards. This is the reason the two methods match to rounding error in Section 5 instead of only looking similar.

3.3 Why a blur kernel blurs

A box or Gaussian kernel is a low pass filter. Its transform $G(u, v)$ is large near the origin (low frequencies) and small far away (high frequencies). Forming $F \cdot G$ keeps the smooth, slowly changing content and cuts the fine detail and sharp edges, which is what blurring means.

4 Worked example by hand

To make the theorem concrete, here is a small case done fully by hand and then checked with the code. Take the length 4 signal and 2 tap filter

$$f = [1, 2, 0, 3], \quad g = [1, 1, 0, 0]. \quad (8)$$

(a) **Direct circular convolution.** Here $y[n] = f[n] + f[n - 1]$ with the indices taken mod 4:

$$y = [1+3, 2+1, 0+2, 3+0] = [4, 3, 2, 3]. \quad (9)$$

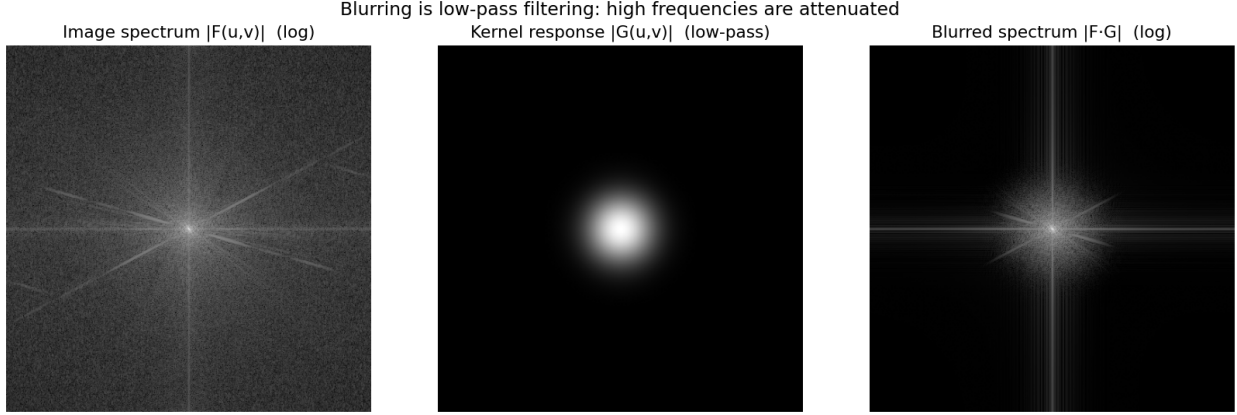


Figure 3: The frequency domain view (log magnitude spectra). Left: the image spectrum. Middle: the kernel response, bright in the centre, which is a low pass filter. Right: the blurred spectrum, with the high frequencies clearly reduced.

(b) **Through the DFT.** Using $\omega = e^{-j2\pi/4} = -j$, so the powers of ω are $1, -j, -1, j$,

$$F = [6, 1+j, -4, 1-j], \quad G = [2, 1-j, 0, 1+j]. \quad (10)$$

Multiplying element by element,

$$F \cdot G = [6 \cdot 2, (1+j)(1-j), (-4)(0), (1-j)(1+j)] = [12, 2, 0, 2]. \quad (11)$$

The inverse DFT of $[12, 2, 0, 2]$ is

$$y = \frac{1}{4} \mathcal{F}^{-1}\{F \cdot G\} = [4, 3, 2, 3], \quad (12)$$

which is the same as the direct convolution in part (a). Multiplying the two transforms and inverting reproduces the convolution exactly.

As a quick 2D check, convolving the patch $[[10, 20, 30], [40, 50, 60], [70, 80, 90]]$ with a normalised 3×3 box kernel gives a centre value of $\frac{1}{9}(10 + 20 + \dots + 90) = 50$, which is just the average of the patch, as expected.

5 Experimental validation

Method. The cameraman image is blurred with the same kernel two ways: the hand written spatial convolution from Section 2, and the Fourier domain multiplication from Section 3. The two outputs are then compared pixel by pixel. If the theorem holds they should be equal apart from floating point rounding.

Panels 2 and 3 of Figure 2 are the two results, and panel 4 is their absolute difference, whose largest value is about 10^{-15} , the scale of machine rounding, so the images are the same. Repeating this across kernels and sizes gives the table below.

Across every setting the biggest pixel disagreement is between 10^{-15} and 10^{-14} , and the PSNR is around 290 to 312 dB. So spatial filtering and its Fourier domain version give the same image, and convolution in space equals multiplication in frequency. The small leftover is not an error in either method; it is the buildup of 64 bit floating point rounding, and it grows slowly with the kernel size (more terms in each sum), which is what you would expect.

Kernel	Size	max abs difference	MSE	PSNR (dB)
gaussian	3×3	2.00e-15	2.14e-31	306.7
gaussian	7×7	1.22e-15	7.11e-32	311.5
gaussian	15×15	2.66e-15	2.23e-31	306.5
gaussian	25×25	3.89e-15	4.02e-31	304.0
box	3×3	1.78e-15	1.63e-31	307.9
box	7×7	1.78e-15	1.03e-31	309.9
box	15×15	6.99e-15	1.66e-30	297.8
box	25×25	1.53e-14	8.31e-30	290.8

Table 1: Difference between the spatial and Fourier results across settings.

6 Repository and how to run

```
Module3_ImageFiltering/
  app.py           Streamlit web app, all parts reachable here
  requirements.txt
  README.md
  src/
    kernels.py     box and Gaussian kernels
    spatial.py     hand written 2D convolution (the blur)
    frequency.py   FFT based filtering (multiply in frequency)
    compare.py     spatial vs frequency experiment and figures
    utils.py       load and save images
    data/sample.jpg built in test image
    outputs/       result figures used in this report
```

Run the web app (this is what the screen recording shows):

```
pip install -r requirements.txt
streamlit run app.py           then open http://localhost:8501
```

Reproduce the experiment and figures from the command line:

```
python -m src.compare --kind gaussian --size 15
python -m src.compare --kind box --size 9
```

Each requirement is shown in the app: blurring on the tab “1. Blur an image”, the spatial equals frequency proof on “2. Spatial = Frequency”, and the theory on “3. Theory” and in Sections 3 and 4 above.

7 References

- R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Chapter 3 on spatial filtering and Chapter 4 on filtering in the frequency domain and the convolution theorem.
- A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, on the DFT and circular convolution.
- NumPy FFT documentation, `numpy.fft`.